SPIKE & LEARN — DESIGN PLAN

One approval page before code: palette, type, layout, the constraint web in every state, and the rules-engine test cases. The court is the hero; everything else stays quiet. Violation red appears in exactly one place on this page — the same discipline the app will keep.

1 · PALETTE — SIX NAMED VALUES

Court maple
#D8A968 — playing surface

Apron maple
#B98A52 — out of bounds, page floor

Court tape
#F6F1E6 — lines, panels, libero jersey

Team navy
#24406B — jerseys, primary actions

Whiteboard ink
#221B12 — text, net, satisfied hairlines

Whistle red
#D7402B — violations ONLY, nowhere else

Legal-formation confirmation uses a desaturated ink-green **#3E6B4F** as a seventh utility value (quiet checkmark, not a celebration).

2 · TWO TYPEFACES

DISPLAY + JERSEY NUMBERS

SAIRA CONDENSED

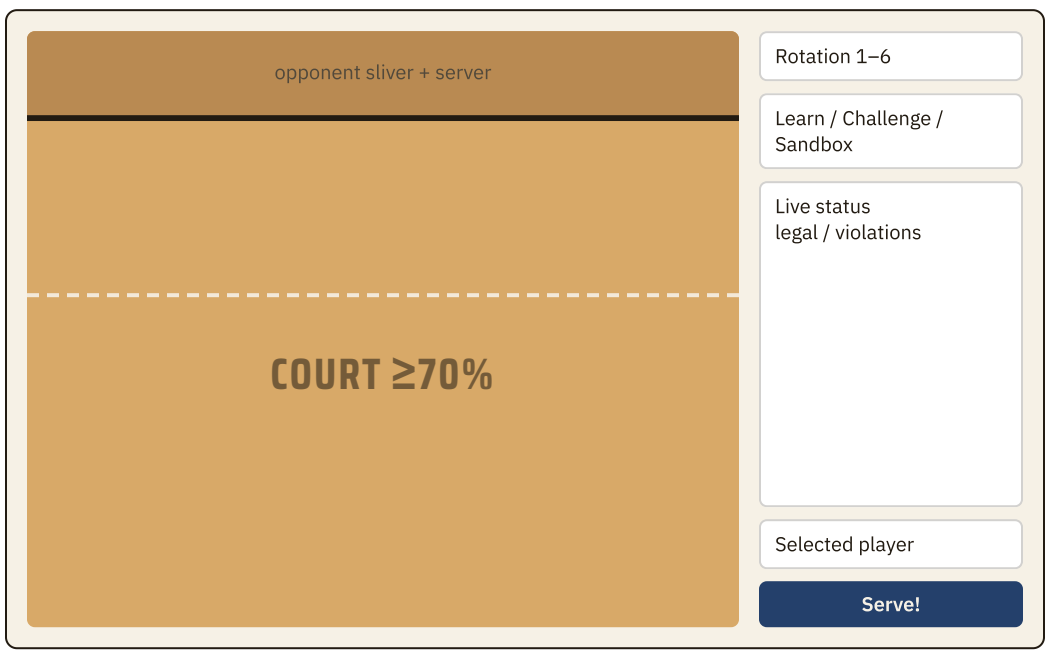
S OH1 L token faces — libero inverted

BODY + RULES TEXT

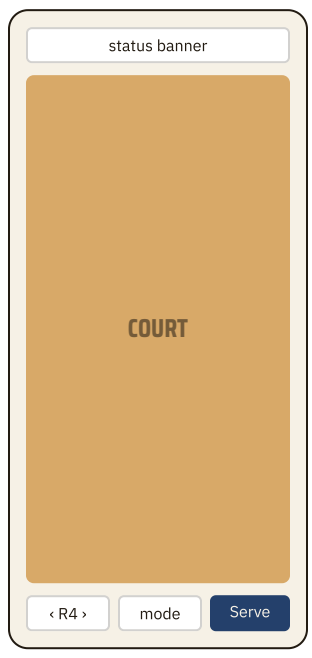
IBM Plex Sans

OPP (zone 6) must stay to the left of OH2 (zone 1). Violation messages name both players and the direction — never “invalid rotation.”

3 · LAYOUT WIREFRAME

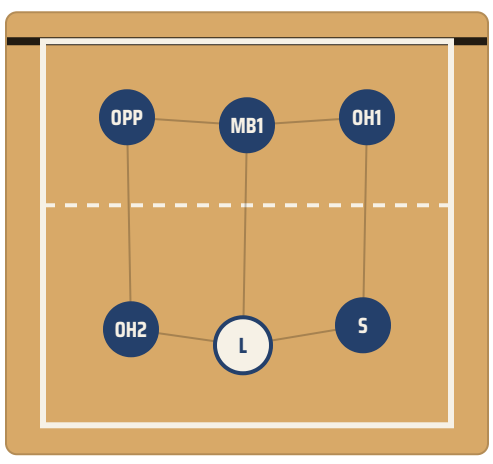


Desktop — court left, one slim rail. No rules wall; “?” opens a compact reference sheet.

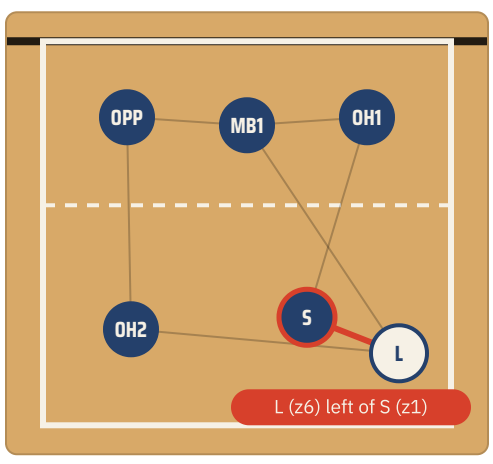


Mobile 360px — bottom bar, scroll locked during drag.

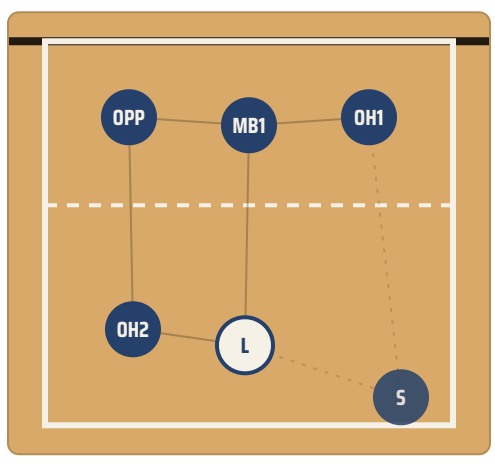
4 · THE CONSTRAINT WEB — THREE STATES



Satisfied — 7 hairlines at 28% ink, almost invisible. Diagonals never drawn: the silent lesson.



Violated — only the broken pair thickens to 3px whistle red, labels itself at the midpoint, rings both tokens. Everything else stays quiet.



Serving (server exempt) — the server’s two lines go dotted-and-faded; the other five keep constraining each other.

Toggle: on by default in Learn and Sandbox, off by default in Challenge. Learn mode additionally shows faint tape-ghost markers (dashed circles, 30% tape white) with the setter’s release arrow, taken from your six rotation diagrams.

5 · RULES-ENGINE SPEC — CONFIRMING YOUR TEST CASES

✓ LEGAL **Rotation 4, stacked setter.** S is in zone 4 — it answers only to zone 3 (MB2, lateral) and zone 5 (OH1, depth). Pressed against the net far left, S is left of MB2 and in front of OH1. Zones 2 and 1 never see the setter.

✓ LEGAL **Rotation 5, OPP tucked mid-court behind the setter.** OPP is zone 6: behind S (3–6 depth), right of L (5–6), left of OH2 (6–1). Mid-court tuck and deep corner both satisfy all three — matching your legal-area diagram.

✓ LEGAL **Zone 5 standing to the right of zone 1.** No 5–1 pair exists among the 7 constraints. No line is even drawn between them.

✗ FAULT **Zone 6 to the right of zone 1.** Breaks the 6–1 lateral pair. Message: “OPP (zone 6) must stay to the left of OH2 (zone 1)” — always both players, role + zone, plain direction.

Engine shape: pure static module. `validate(positions, {servingTeam}) → {legal, violations:[{pair:[a,b], axis, message}]}`. Exactly 7 pairs: 4–3, 3–2, 5–6, 6–1 lateral; 4–5, 3–6, 2–1 depth. Server (zone 1, our serve) excluded from all pairs. Comparisons strict (<), positions continuous, judged only at the “Serve!” snapshot — live feedback simply runs the same function every pointer move.

Also modeled: per-rotation role map from the serving order S→OH1→MB1→OPP→OH2→MB2; libero auto-swap for the back-row middle; coach’s-answer formations digitized from your six SVGs; post-serve release animation to playing positions.

Reply “go” to build, or flag anything above. Buildable tweaks later: team color, court tone, web thickness — exposed as tweak props.